

Predicting Vehicle Loan Default

Classification using the CRIF High Mark vehicle loan dataset

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Data and Problem

Data

- CRIF High Mark vehicle loan dataset
- $n = 233,154$
- Thin file: XXXX
- Normal: YYYY

Method

- LOREM IPSUM

Tools

- R: for EDA, cleaning, and modelling

Questions of Interest - maybe this is methods?

Model 1: Random forest

Model 2: XGBoost

Findings: Random Forest Method

Findings: XGBoost Method

Obstacles: Feature Engineering

The high dimensionality dataset had several complicated aspects to compensate for

- Dimensionality Reduction
- High Cardinality Variables
- Creating stress score

Stress score image

Obstacles: Imbalanced Categories

Higher Proportion of Defaulted Observations

- Different models performed better on a specific category
- XGBoost parameter adjustments
- SMOTE Methods

Future Work

Several avenues for additional work

Area 1: Splitting Dataset by thin file and training two models

Area 2: TBD